

Khelan Mehta

AI Engineer · ESG & Sustainability Automation · RAG & Agentic Systems

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AI engineer who ships production LLM systems **and** understands the sustainability reporting they have to serve. Designs and ships the tooling that removes manual reporting work: **Verdantra**, a deployed ESG platform that collects data once and reports to GRI, BRSR, ESRS/CSRD, SBTi and ISSB with page-cited provenance, and **Pinocchio**, a cloud DOE-2.3 building-energy copilot answering questions from real simulation runs. Built the AI platform at CannonDesign that collapsed an **80-hour energy-model documentation workflow into ~2 hours**. Hands-on across the full RAG stack — hybrid retrieval, reranking, agentic pipelines, evaluation and hallucination control. LEED AP BD+C. Winner of 5+ national hackathons, including 2nd of 2,200+ teams at IEEE Tic Tech Toe 2026.

AUTOMATION PLATFORMS — DESIGNED, BUILT & DEPLOYED

Verdantra — ESG Reporting Automation Platform verdantra.in

Deployed · 2026

Sole engineer — end-to-end build — automating the ESG data-collection and disclosure cycle for reporting teams

- **The automation:** ESG reporting is a structured-data problem in a document's clothing. Verdantra replaces the per-client, per-framework spreadsheet cycle with a canonical **framework-neutral metric store** (MongoDB) plus a **disclosure crosswalk** — one collected data point maps to GRI, BRSR, ESRS/CSRD, SBTi and ISSB at once. **Collect once, report to many.**
- Built a **continuous gap-analysis engine** that reproduces the manual QA/QC review: it scores disclosure coverage by E/S/G pillar, ranks findings blocking → minor, and holds publication on unresolved conflicts — surfacing, for example, a diesel volume in a client workbook that contradicts the Scope 1 figure in their GHG inventory.
- Engineered the grounding layer so **no number reaches a report without provenance** — every consequential agent action is a tool call against stored data, and every claim carries a file-and-page citation back to raw evidence. Audit-readiness by construction.
- **Voice agent ("Vera")** as the single interface — teaches the framework, collects inputs by speech or upload, clarifies conflicts and answers queries in any auto-detected language; a vector store handles the ~40% of ESG data that is unstructured (policy PDFs, inventories, diagrams).
- Validated against a real FY2025 engagement: the engine reproduced the consultant's hand-built gap deck and independently surfaced the reconciliation conflict a manual line-by-line review would hunt for. Stack: MongoDB · Qdrant · Redis · self-hosted BGE-M3 embeddings · OpenAI (LLM + speech) · Cloudflare, on a free-tier-first architecture with near-zero marginal cost per engagement.

Pinocchio — Building-Energy Simulation Copilot pinocchio-app.vercel.app

Deployed · 2026

Sole engineer — end-to-end build — automating the energy-model edit-run-read loop for modelling teams

- **The automation:** energy modellers answer questions by hand-editing a model, re-running it and reading the output. Pinocchio puts a **real DOE-2.3 engine** behind a copilot — questions are answered from **actual simulation runs, not model recall**, with parameter knobs and an equipment inventory that trigger re-simulation on change.
- Built **optimize_to_target**, a stacked real-engine optimizer that searches parameter sets against an energy target, collapsing the manual edit-run-read loop into a single instruction; Results and Inventory dashboards render each run.
- Shipped a **/train knowledge console** that ingests existing simulation models as training projects, so the copilot grounds its answers in the firm's own model corpus rather than generic assumptions.
- Full-stack ownership: React + Vite SPA with a three.js landing on Vercel; Express API and simulation engine containerised on Fly.io; MongoDB Atlas; self-hosted **JWT auth** (bcrypt) with token-authenticated **SSE streaming** for the chat transport.

EXPERIENCE

AI Engineer, Sustainability Automation (*Intern*)

May 2026 – Jul 2026

CannonDesign — Mumbai, India

- Architected and shipped **The Marcus Studio**, a full-stack AI platform parsing eQUEST, TRACE 3D Plus and IES-VE energy-model outputs (.SIM / .inp / gbXML) into a single unified schema — **cutting an 80-hour manual BIM-to-documentation workflow to roughly 2 hours**. Co-presented the platform internally with my manager.
- Built the platform's four pillars: a **QAQC dashboard** flagging model inconsistencies pre-review, **MEPC calculator automation**, **utility-rate intelligence**, and an **AI search layer** over the parsed model corpus.
- Engineered the LLM/RAG tooling behind it — retrieval design, chunking strategy over semi-structured simulation output, and grounded generation into structured LEED documentation with traceability back to source.
- Delivered ASHRAE 90.1 and Title 24 compliance modelling for LAUSD K-12 projects via Revit-to-gbXML geometry workflows and Rhino/Grasshopper parametric analysis.

Automation & Sustainability Engineer

Jun 2025 – Dec 2025

Ergo Energy LLP — Surat, Gujarat

- Designed an **ESG KPI tracker and disclosure-framework mapper** — mapping reporting requirements down to underlying data points, tracking KPIs against the correct framework, and dashboarding gaps and missing evidence ahead of assurance.

- Solved the hard part of ESG tooling: reconciling one messy set of source data against multiple disclosure structures **without losing traceability** from disclosure back to raw evidence.
- Built Python automation pipelines processing simulation outputs into structured reports, **cutting analysis time 40%** and eliminating manual data entry; supported 3 LEED certifications (BD+C, O+M) with baseline energy models and performance documentation.
- Ran 8+ building energy audits with ECM recommendations and ROI analysis, identifying **15-20% annual energy cost savings**; shipped the firm's client-facing platform (ergoenergysolution.com).

Automation Engineering Intern May 2026
LEAD Consultancy and Engineering Services (India) Pvt Ltd — Bengaluru, Karnataka

- Built **workflow automation** for engineering consultancy processes, replacing manual, repeat-heavy steps with scripted pipelines.

Development Team Manager Jun 2023 – May 2024
The IT Company — Brown Ion — Remote

- Led a **6-developer team** delivering 10+ production web applications on agile cycles, shipping **20% ahead of schedule** at 99.9% uptime; architected MERN systems to OWASP standards and built ML models automating internal workflows, reducing operational cost 10%.

Full-Stack Engineer Aug 2021 – Jun 2023
Admyre — Remote

- Built an influencer marketing platform (React / Node / Express) connecting **20+ brands** to creators, processing \$2K+ in campaign transactions, with dashboards tracking 300+ engagement metrics across 15+ monthly campaigns.

COMPETITIVE WINS — 5+ NATIONAL HACKATHONS

2nd / 2,200+ **Tic Tech Toe 2026 (IEEE)** — 1st in track. **Skilleton**, a two-layer workforce-intelligence system scraping India's live job market into city-matched reskilling paths. Owned platform & DevOps: Express backend, JWT-auth public REST API across 12 route namespaces, Docker Compose deployment, OpenAPI docs.

1st **IEEE DSC 2.0 Hallucination Detection Challenge** — **HalluciNot**, a Flask pipeline detecting and mitigating hallucinations in LLM output — the exact failure mode that makes RAG unusable in audit contexts.

1st **Mitraroop Startup Hackathon 2026** — **NoteNex**, an AI notes platform with smart summarization, tagging and knowledge organization. **1st** **GDG Techsprint 2025** — **Cookie**, a real-time animal emergency response platform matching rescuers to urgent cases.

4th / 1,600+ **HackaMined 2026** — 1st in track, across 150+ universities, CSI Nirma University.

SELECTED PROJECT

equestRag — AI Document Analysis System github.com/khelan-mehta/equestRag

- RAG architecture over **1,000+ building energy documents** using FAISS embeddings and semantic search, with 95% extraction accuracy across PS-E, BEPU and LS-C report formats; ML benchmarking against ASHRAE at 92% accuracy.
- NLP query interface handling 500+ natural-language questions on EUI, HVAC loads and performance; automated PDF reporting saved 15 hours weekly.

TECHNICAL SKILLS

ESG & Reporting	ESG KPI tracking, disclosure-framework mapping & crosswalks, GRI / BRSR / ESRS-CSR / SBTi / ISSB, materiality assessment, carbon accounting & GHG inventory reconciliation, LCA, gap analysis & audit-readiness workflows, ESG dashboarding
AI / LLM	RAG (chunking, hybrid retrieval, reranking, evaluation), agentic & multi-step pipelines, tool/function calling, hallucination detection, grounding & citation provenance, FAISS / Qdrant, embeddings (BGE-M3), voice agents, prompt engineering, TensorFlow, PyTorch, scikit-learn, NLP
Energy Modeling	eQUEST, DOE-2.3, IES VE, TRACE 3D Plus, EnergyPlus, ASHRAE 90.1, Title 24, LEED BD+C & O+M, Revit, Rhino/Grasshopper, gbXML
Engineering	Python, FastAPI, Flask, JavaScript / TypeScript, Node.js, Express, React, Next.js, PostgreSQL, MongoDB, Redis, Docker, Fly.io, Vercel, Cloudflare, REST / OpenAPI, SSE, JWT auth, OWASP, Git

CERTIFICATIONS

LEED AP Building Design + Construction — USGBC, Dec 2025
LEED Green Associate — USGBC, Aug 2025
CSR Fundamentals Level 1 — CSR Institute, Sep 2025
ESG Performance Measurement — Alison, Sep 2025
Life Cycle Assessment (Beginner) — Ecochain, Sep 2025

EDUCATION

B.Tech, Electronics & Communication Engineering
Nirma University, Ahmedabad — 2022-2026 · CGPA 8.12/10
Data Structures & Algorithms · Machine Learning · Computer Networks · Operating Systems
Higher Secondary (CBSE) — 91.6%
Essar International School, Surat — 2020-2022 · JEE Mains 93.53%ile